THEORETICAL PHYSICS · CONSCIOUSNESS STUDIES · QUANTUM FOUNDATIONS

THE SILENCE EXPERIMENT

A Multi-State Protocol for Detecting
Consciousness-Dependent Decoherence in Shielded Environments

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PAPER 2 of 2

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SECTION I

Abstract

This paper proposes a controlled experimental protocol for testing the von Neumann-Wigner (VNW) interpretation of quantum mechanics, the hypothesis that consciousness plays a role in quantum state reduction, against the prevailing environmental decoherence framework (Zurek, Zeh, Joos). The protocol measures whether the cognitive state of a nearby conscious observer measurably affects the decoherence rate of a prepared quantum system under controlled conditions.

The experimental platform places an experienced meditator in EEG-verified states ranging from active cognition to objectless awareness inside a shielded environment. A quantum coherence sensor measures decoherence rates (T2) across seven controlled consciousness conditions (C1 to C7) spanning the full cognitive-excitation axis, against an empty-chamber instrumental baseline. The primary pre-registered hypothesis is that T2 at C7 (Cessation / Objectless Awareness) and T2 at C6 (Minimal Phenomenal Experience) will both exceed T2 at the chamber baseline and at conditions C1 to C5, and that T2 at C7 will further exceed T2 at C6. This is tested via two pre-registered orthogonal planned contrasts: (i) [C6, C7] vs. [baseline, C1-C5], and (ii) C7 vs. C6, both with directional predictions. All other cross-condition comparisons are treated as exploratory. The multi-state design allows the data to discriminate between a single-point cessation effect, a graded depth-of-practice effect, and a null.

A secondary experiment uses quantum random number generators with improved shielding, blinding, and instrumentation.

Phase 0 is a foundational test of the von Neumann-Wigner interpretation under controlled observer states, a question that has been part of the quantum literature for nearly a century without ever being tested this way. The chamber's specifications were dictated by what is required to make any consciousness-dependent T2 signal interpretable: drive every known environmental coupling channel below the noise floor and monitor each one synchronously. A sensor-and-shielding characterisation of the NV-center platform under that isolation regime, with quantified upper bounds on residual electromagnetic, thermal, vibrational, and magnetic coupling to T2, is produced as an independently publishable co-product of the same hardware run, in a metrology venue. The chamber exists for the consciousness test; the metrology paper is what falls out of doing the consciousness test correctly.

The protocol is phased: a Phase 0 study (~$451,200; range $420,000 to $470,000) using nitrogen-vacancy centers in diamond, and a Phase 1 full protocol (~$1M to $3M) using superconducting qubits in an underground laboratory, conditional on Phase 0 results and adoption by an institutional host. (A small commercial-QRNG exploratory activity is described separately in Appendix A as personal exploratory activity, not part of the main protocol.) Analysis is pre-registered, randomised, outcome-blinded where feasible, and fully blinded by an independent statistician who does not see condition labels until the analysis is locked. It is not designed to test small-effect claims from prior consciousness-RNG literature.

This protocol is proposed as a contribution to an ongoing line of experimental inquiry into the measurement problem in quantum mechanics, not as a definitive resolution. Its value lies in the

specificity of its predictions and the rigor of its controls. Null results establish pre-registered upper bounds; positive results require independent replication.

SECTION II

Introduction: The Measurement Problem and the Observer

Quantum mechanics contains an unresolved foundational question known as the measurement problem. A quantum system exists in superposition, multiple states simultaneously, until it undergoes a transition to a single definite outcome. What causes this transition has been debated since the formalization of quantum theory in the late 1920s and remains unresolved today.

Every major interpretation of quantum mechanics is, at its core, a different answer to this question. The Copenhagen interpretation treats measurement as a primitive concept without further analysis. The Many-Worlds interpretation (Everett, 1957) denies that collapse occurs at all. Bohmian mechanics introduces hidden variables. QBism treats quantum states as subjective degrees of belief. And the von Neumann-Wigner (VNW) interpretation, first articulated by von Neumann in 1932 and extended by Wigner in 1961, proposes that consciousness itself is the agent of state reduction.

The VNW interpretation is not mainstream. It is, however, not empirically refuted. Its marginal position in contemporary physics reflects philosophical discomfort, the success of environmental decoherence theory, and the absence of controlled experiments designed to distinguish between interpretations using the observer's cognitive state as an experimental variable.

2.1 Environmental Decoherence: The Mainstream Position

The environmental decoherence program (Zurek, 2003; Zeh, 1970; Joos et al., 2003) provides a mechanism by which quantum superpositions become effectively classical through interaction with the environment. However, decoherence does not solve the measurement problem in its entirety. As Zurek has acknowledged, decoherence explains the suppression of interference but does not explain why a specific outcome is realized from among the surviving diagonal elements. This residual gap, the problem of definite outcomes, is where the VNW interpretation operates.

2.2 The Boundary Effect

The theoretical framework proposes a boundary effect: decoherence is the quantum-classical boundary, and if consciousness participates in state reduction, the effect should manifest at this boundary as a shift in coherence times. This experiment measures not consciousness itself but the shift in the decoherence boundary associated with the observer's state.

2.3 The Experimental Gap

Controlled experiments systematically varying the cognitive state of a conscious observer while measuring the decoherence rate of a nearby quantum system represent a gap in the existing literature in quantum foundations. Decoherence experiments routinely characterise T2 against electromagnetic, thermal, vibrational, and magnetic perturbations; none have included the observer's cognitive state as a controlled independent variable verified by EEG. This protocol aims to address that gap, with improved methodology, more precise state characterization via EEG, and a decoherence-focused measurement strategy.

SECTION III

Theoretical Context

3.1 The Von Neumann-Wigner Interpretation

In his 1932 treatise, von Neumann demonstrated that the boundary between the quantum system and the measuring apparatus can be moved arbitrarily far along the measurement chain without affecting the predictions of the theory. The logical terminus of this chain is the conscious observer. Wigner (1961) made the stronger claim explicit: consciousness is necessary for quantum state reduction. Stapp (2007) developed this view further. The present experiment tests a weaker, empirically tractable version: if consciousness participates in state reduction, then varying the cognitive state of a nearby conscious observer should produce measurable differences in decoherence rates.

3.2 Environmental Decoherence Theory

Environmental decoherence theory predicts that the cognitive state of a nearby conscious observer should have no effect on the decoherence rate of a quantum system, provided environmental coupling remains constant. This is the null prediction the experiment is primarily designed to test.

3.3 Orchestrated Objective Reduction (Orch-OR)

Penrose and Hameroff (2014) proposed that consciousness arises from quantum processes in neuronal microtubules, with gravity triggering collapse at a threshold. Orch-OR makes distinct predictions from VNW, though quantitative predictions for this specific experimental configuration have not been derived. This is a limitation.

3.4 Neuroscientific Basis for Meditation as a Controlled Variable

Deep meditation produces neurologically distinctive states well-documented in the peer-reviewed literature. Lutz et al. (2004) reported elevated gamma-band power and long-range phase synchrony in long-term meditators. Berkovich-Ohana et al. (2012) documented EEG correlates of mindfulness-induced changes in gamma band activity, including DMN deactivation. Winter et al. (2020) provided further characterization of neural markers associated with content-free awareness states. Metzinger (2020) and subsequent work on Minimal Phenomenal Experience (MPE) operationalises the threshold state of tonic alertness without intentional content as phenomenologically and neurophysiologically distinct from both ordinary resting awareness and full cessation, providing the construct basis for Condition C6 in this protocol.

Key features include: gamma wave coherence, default mode network suppression (Brewer et al., 2011), distinct differentiation from sleep, and structural brain changes. EEG identifies neural correlates consistent with these states, not the subjective states themselves. EEG provides convergent evidence for the target state, not definitive verification of subjective experience.

3.5 T2 as an Operational Proxy

The VNW interpretation concerns wavefunction collapse in a measurement context, while T2 measures the loss of off-diagonal coherence in a quantum system's density matrix. These are not the same process, and a null on T2 does not rule out a VNW effect in pure measurement-outcome statistics. Decoherence time is used here as an operational proxy for any putative consciousness-coupling channel, not as the canonical VNW observable. The reasoning is that if consciousness modulates state reduction in any way, the most accessible empirical signature would be a shift in the rate at which superpositions decohere into classical mixtures, detectable as a change in T2. A null result therefore constrains but does not eliminate the consciousness-causation hypothesis.

SECTION IV

The Mechanism Problem

The VNW interpretation provides no mechanism by which consciousness affects quantum systems. This is its primary theoretical weakness and the principal reason for its marginal position. The experiment as designed is genuinely agnostic on mechanism: it asks whether the cognitive state of a nearby conscious observer correlates with measurable differences in T2, not how. If a robust correlation is detected and replicated, that finding itself becomes the empirical anchor for mechanism research; if no correlation is detected, the null result constrains the parameter space within which a VNW-style coupling could exist.

The protocol does not assume that consciousness emits an electromagnetic signal, a gravitational signal, or any other known field. Such a signal would be detectable by ordinary instruments and is independently characterised in the chamber's environmental monitoring array (see Section 5.4). If a consciousness-correlated T2 shift is observed in the absence of any environmental signature, the shift itself becomes the empirical phenomenon to be explained, by whatever theoretical framework subsequent work establishes.

This agnosticism on mechanism is deliberate. The history of physics contains many phenomena whose detection preceded their explanation by decades (e.g., the photoelectric effect, the cosmic microwave background). The present protocol is structured to detect a phenomenon, not to prove a mechanism.

SECTION V

Experimental Design

5.1 Multi-State Consciousness Conditions

The independent variable is the observer's cognitive state, operationalised as seven controlled conditions spanning the full cognitive-excitation axis from maximum (C1) to complete cessation (C7), against an empty-chamber instrumental baseline (no observer present). Each condition is defined by a specific cognitive instruction, expected phenomenology, and EEG signature.

Cond.	Name	Cognitive Instruction	Expected EEG Signature
Baseline	Empty chamber, no observer	All instrumental noise sources only	Instrumental control
C1	Active cognition	Continuous mental arithmetic / verbal task	Increased beta and broadband gamma; active frontoparietal network engagement
C2	Ordinary resting state	Eyes closed, awake, no instruction	Alpha-dominant resting EEG with normal default mode network activity
C3	Focused attention (Shamatha)	Single-pointed concentration on breath	Increased beta and gamma coherence; frontal midline theta
C4	Open monitoring (Vipassana)	Receptive awareness without specific focus	Increased theta; decreased beta; characteristic parietal changes
C5	Non-directed openness	Complete receptive surrender without technique	High alpha/theta coherence; heart-brain coherence; sustained low beta
C6	MPE / Luminous bare awareness	Tonic alertness without intentional content; no active self-model or object	Collapse of frontoparietal beta; sustained low-frequency alpha/theta with high spectral purity
C7	Cessation / Objectless awareness	Contentless awareness; full cessation (nirodha) where attainable	Gamma burst then flattening, or sustained high-amplitude gamma; profound suppression across frequencies

Table 1: The seven consciousness conditions and the empty-chamber baseline. Conditions are listed in descending order of cognitive excitation.

The conditions form a structured axis rather than a pair of contrasts. C6 (MPE) is the threshold state immediately preceding full cessation; it is included as a distinct condition because the contemporary phenomenological and neurophysiological literature (Metzinger 2020 and subsequent) treats it as a discrete state of tonic alertness without intentional content, not merely a transition. C7 (Cessation) is the deepest accessible state and the primary site of the experiment's pre-registered prediction. The other conditions (C1 to C5) provide the gradient against which any C6/C7 effect is tested.

Note on C6 (MPE): C6 is included as a discrete pre-registered condition not because it is hypothesised to produce the largest effect, the primary prediction places that at C7, but because it is the operationally cleanest state on the cognitive-excitation axis. C7 (cessation) is, by its phenomenology, a state from which the observer cannot reliably initiate or terminate measurement; C6 (MPE), by contrast, has a stable EEG signature, can be sustained on instruction, and is the contemporary literature's clearest operationalisation of 'consciousness without content.' Including C6 as a separate pre-registered cell allows the protocol to test the threshold-vs-cessation contrast (C6 vs. C7) directly, which is structurally more informative than collapsing C6 into either C5 or C7 would be.

Verification of state attainment relies on the EEG signatures defined above as the primary objective marker for each condition, supplemented by participant self-report following each session. Self-report is recorded for protocol completeness and quality control but is not used as a primary criterion for inclusion in the C6 or C7 analysis cells. C6 specifically requires real-time EEG verification of MPE-consistent signature (collapse of frontoparietal beta, sustained low-frequency alpha/theta with high spectral purity) before any trial is included in the C6 cell. C7 inclusion requires both prior demonstrated reliability of cessation entry under laboratory conditions and EEG patterns consistent with the target signature during the recording window; sessions failing either criterion are excluded from C7 analysis cells per pre-registered rules and are analyzed under their actual EEG-classified condition.

5.1.1 EEG Classification of C6/C7 and the Pilot Firewall

The contemporary literature does not yet provide a single, replicated EEG signature that uniquely identifies MPE (C6) or cessation (C7); the descriptions in Table 1 are target characterisations, not a validated classifier. Because EEG verification is the inclusion criterion for C6 and C7, this protocol pre-registers a positive, quantitative classifier rather than relying on a qualitative match. The classifier specifies named channels, named frequency bands, explicit amplitude/coherence thresholds, and a defined scoring window for each of C6 and C7. It is defined positively (what the state is), never by exclusion ('anything not matching C1-C5'), so that it is falsifiable: a deviant trace that fails the C5 criteria is not thereby classified as C6 or C7.

Pilot firewall. Exploratory pilot recordings may be used to develop and anchor the classifier thresholds, cross-checked against the practitioner's first-person report of state attainment. Once specified, the classifier is frozen and lodged in the pre-registration before Phase 0 data collection begins. Pilot data are never pooled into the confirmatory Phase 0 dataset, never used to set the empty-chamber baseline, and (pool size permitting) pilot participants are excluded from the confirmatory sample, so that the criterion is not trained and tested on the same data. Trials whose recorded EEG fails the frozen C6 or C7 classifier are excluded from those cells and analysed under their actual EEG-classified condition, per the rules in Section 5.6 and the stopping rules in Section X.

5.2 The Shielded Chamber and Five-Layer Isolation

The chamber is the engineering foundation that makes any consciousness-arm signal interpretable. Its specifications are not chosen for convenience or cost; they are dictated by what is required to drive every known environmental coupling channel below the threshold at which it could plausibly account for a T2 variation across consciousness conditions. The full specification is given in the companion engineering blueprint; the table below summarises the key parameters.

Parameter	Specification
Inner volume	3.0 m x 3.0 m x 3.0 m perfect cube
Layer 1 (innermost)	Acoustic foam + aerogel + Peltier-cooled active liner; +/- 0.01 K thermal stability
Layer 2 (magnetic)	Dual-layer mu-metal (2a inner + 2b outer) with 50 mm air gap; > 80 dB DC and AC (50/60 Hz)
Layer 3	OFHC copper, welded seams; > 100 dB RF attenuation (10 kHz to 10 GHz)
Layer 4	Pneumatic isolators + neoprene dampers; < 10^-6 m/s^2 at sensor platform (1 to 100 Hz)
Layer 5 (outer)	Aluminum structural shell + copper mesh Faraday cage
Total EM attenuation	> 120 dB combined across 10 Hz to 10 GHz
Access	Dual-door airlock with RF / fluxgate contamination check
Reverification	Annual full re-commissioning; post-relocation, post-mechanical-shock, and post-panic-bar invalidation procedures

Table 2: Chamber summary specifications. Full engineering detail is given in the companion engineering blueprint.

The chamber design has the structural property that any residual variation in T2 across consciousness conditions cannot plausibly be attributed to environmental coupling, because the relevant channels have been driven below the sensor's intrinsic T2 limit by at least an order of magnitude.

5.3 Quantum Sensor Options

Parameter	Option A: Superconducting Transmon Qubit	Option B: NV-Center in Diamond
T2 (decoherence time)	50 to 200 microseconds	1 to 10 milliseconds at room temperature
Operating temperature	~15 mK (dilution refrigerator)	Room temperature
Cost (sensor + readout subsystem)	~$400K to $800K (full platform incl. integration and shielding)	~$30K to $80K (full platform incl. integration)
Sensitivity	Highest sensitivity to environmental perturbation	Lower sensitivity, longer baseline T2
Advantages	Maximum detection sensitivity	Cheaper, simpler, no cryogenics
Disadvantages	Expensive, complex, requires specialized operators	Less sensitive to small perturbations

Table 3: Quantum sensor options for decoherence measurement. Cost ranges shown are full sensor-platform figures including optical table, integration mounting, calibration overhead, and platform-level shielding/vibration decoupling. The corresponding line items in the engineering Bill of Materials (NV sensor kit at ~$17,550; transmon cryostat alone at $234,000 to $585,000) cover the sensor module only; the difference is integration and platform infrastructure, budgeted under Structural and Labor. Full Phase 0 and Phase 1 budgets in Section XI include shielding, support systems, and labor across all categories.

Recommendation: NV-center in diamond for Phase 0. If a positive signal is detected and replicated, upgrade to superconducting qubit for Phase 1.

5.4 Instrument Array and Environmental Monitoring

Instrument	Measures	Target	Sensitivity
Quantum Coherence Sensor (NV or transmon)	Decoherence rates (T2)	Primary experiment	<= 1 us resolution
QRNGs (x4)	Quantum randomness statistics	Secondary experiment	>= 1 Mbit/s, quantum-sourced
64-Channel Wireless EEG	Neural state characterization	Independent variable	>= 1 kHz sampling
SQUID Magnetometers	Magnetic fields (femtotesla)	Environmental monitor	< 10 fT/sqrtHz (LTS, 4 K)
Single-Photon Detectors	Optical noise / biophotonic signals (exploratory)	Exploratory; not pre-registered	< 100 dark counts/s
3-axis fluxgate magnetometer	Residual magnetic field	Environmental monitor	< 10 nT resolution
3-axis seismic accelerometer	Vibration at sensor platform	Environmental monitor	< 10^-6 m/s^2 (1 to 100 Hz)
Pt100 RTD sensors (x4)	Temperature drift	Environmental monitor	+/- 0.001 K
Scintillator coincidence muon veto	Cosmic-ray muon arrival times (paired-panel coincidence)	Environmental monitor (Phase 0)	Flags muon-coincident T2 events for exclusion
Acoustic / pressure / humidity	Acoustic SPL, baro, humidity	Environmental monitor	Standard lab-grade

Table 4: Instrument array and environmental monitoring channels. All channels recorded synchronously via GPS-disciplined 10 MHz common time base (< 1 us inter-channel synchronization).

If any anomaly appears in quantum data, environmental channels confirm whether it correlates with a physical artifact or is specific to the consciousness condition. This is the most critical step in the artifact analysis.

Observer physiological covariates. Because the observer's body is the one coupling source inside the shielded volume, a set of observer-physiology channels is recorded synchronously with all other channels (same GPS-disciplined time base) and pre-registered as nuisance covariates in the analysis model (Section 8.5): respiration (chest/abdomen belt), electrocardiogram (heart rate and variability), surface electromyography (gross muscle tone and micro-movement), and a body-directed infrared/thermal channel (metabolic heat and skin temperature). These variables change systematically across the cognitive-excitation axis (slower respiration, reduced movement, and lower thermal output at C6/C7), so monitoring them directly converts the 'the deep states simply involve a stiller body' objection into a testable, pre-registered control rather than an unmeasured confound. A consciousness-linked T2 effect is credited only if it survives adjustment for these somatic covariates; respiration and heart rate are treated as partially constitutive of the deep states and are reported both as covariates and in a sensitivity analysis without adjustment, so the somatic and neural contributions can be distinguished rather than conflated.

5.5 Session Protocol

All sessions follow a strict single-observer rule.

5.5.1 Garment and contamination control

Observers wear only facility-provided non-metallic garments. No jewelry, conductive materials, synthetic fibers, or external devices are permitted inside the chamber. All personal items are removed and stored in the external preparation area prior to airlock entry.

5.5.2 Airlock contamination check

Upon entering the airlock, the observer remains for a 60-second stabilization period during which an RF power meter confirms residual electromagnetic emission below -80 dBm (10 kHz to 10 GHz) and a fluxgate magnetometer confirms no ferromagnetic contamination. Only after the contamination check passes is the inner door released.

5.5.3 Strict single-observer rule

Only one observer is permitted inside the main cubic volume during any active data-collection session. No assistants, additional personnel, or equipment beyond the pre-installed sensor array are present.

5.5.4 Session timing

Session timing is harmonised across phases: every consciousness condition has the same duration in Phase 0 and Phase 1. A 5-minute pre-condition settling buffer precedes each consciousness condition. C1 to C6 each have a 15-minute condition period; C7 (cessation) has a 30-minute condition period. The structural difference between phases is preserved: Phase 0 runs one condition per session; Phase 1 chains all seven conditions in a single randomised session.

Phase 0 (single-condition sessions): Each session consists of one consciousness condition only. Structure: empty-chamber calibration (10 min, before observer ingress), then observer entry and ingress settling, then a 5-minute pre-condition settling buffer, then the condition period (15 min for C1 to C6, 30 min for C7), then a brief post-session baseline. No inter-condition buffers are required within a session because no condition transitions occur. One session per observer per 24 h. Empty-chamber baseline measurements are run as separate sessions on the same randomised schedule.

Phase 1 (chained multi-condition sessions): A single session runs all seven consciousness conditions in randomized order, with the empty-chamber baseline measured in dedicated separate sessions. Condition periods are 15 minutes each for C1 to C6 and 30 minutes for C7. A 5-minute pre-condition settling buffer precedes each condition. Total in-session condition plus buffer duration: 6 x (5 + 15) + 1 x (5 + 30) = 155 minutes, plus empty-chamber calibration (10 min, pre-ingress) and ingress settling, yielding approximately 3 hours of total session length. C7 is positioned last in the condition order whenever the randomization permits, to avoid placing the four-step exit protocol (Section 5.5.6) mid-chain. One session per observer per 24 h.

Rationale for harmonised timing: fatigue and habituation effects scale with total session occupancy, not with phase. Identical block structure across phases simplifies cross-phase comparison, eliminates a degree of freedom in the protocol, and is consistent with the principle that the cognitive condition itself, not the time allocated to it, is the experimental variable. The 30-minute window for C7 reflects the structural requirement that cessation states are sustained, not entered-and-exited rapidly; the 15-minute window for C1 to C6 is sufficient for stable EEG-verified entry and a representative data-collection sample.

5.5.5 Emergency procedures

Inner-door panic bar allows immediate egress. Fiber-optic video and intercom monitoring throughout. At no point during normal operation are both airlock doors open simultaneously.

5.5.6 C7 exit protocol

Condition C7 (cessation) is structurally distinct in requiring a dedicated four-step exit procedure: gradual re-engagement of intentional content, return of self-model, return of discursive cognition, and verbal check-in via fiber-optic intercom. The exit procedure protects the observer and is logged but is not part of the C7 data window.

5.6 Participants

- C3, C4, C5, C6, C7: minimum 5 experienced meditators with documented ability to reliably produce EEG patterns consistent with the target states (gamma coherence + DMN suppression for C6/C7; condition-specific markers for C3 to C5) under laboratory conditions.
- C6 (MPE) specifically: participants must demonstrate reliable entry into the MPE threshold state with stable EEG signature (collapse of frontoparietal beta, sustained low-frequency alpha/theta with high spectral purity) prior to data collection. Self-report alone is insufficient.
- C5 (Non-Directed Openness): minimum 3 experienced contemplative practitioners with documented practice in non-directed receptive openness.
- C2 (Untrained Resting): minimum 5 non-meditators matched for age and sex.
- C1 (Active Cognition): same meditators as C3 to C7 (within-subjects comparison).

5.7 Blinding and Replication

- Outcome blinding: the T2 analysis pipeline produces decoherence estimates from raw quantum sensor data without access to condition labels. Condition labels are not joined to T2 values until the pre-registered analysis is locked.
- Independent statistician: a statistician outside the experimental team performs the locked analysis. They do not see condition labels during pipeline development.
- Pre-registration: the full analysis pipeline, hypotheses, contrasts, and stopping rules are pre-registered at OSF before data collection begins.

- Replication requirement: any positive result requires independent replication by a non-affiliated laboratory before being treated as established.
- Sampling plan: multiple sessions per participant, clustered; analyzed per Section 8.5, power per Section IX.

SECTION VI

Primary Experiment: Quantum Decoherence Test

The primary experiment measures T2 (the transverse coherence time) of the quantum sensor across the seven consciousness conditions and the empty-chamber baseline. Each condition contributes a population of T2 measurements; the cross-condition comparison is the test of the hypothesis.

6.1 Primary Pre-Registered Hypothesis

T2 at C7 (Cessation / Objectless Awareness) and T2 at C6 (Minimal Phenomenal Experience) will both exceed T2 at the chamber baseline and at conditions C1 to C5. Further, T2 at C7 will exceed T2 at C6. This is tested via two pre-registered orthogonal planned contrasts, both with directional predictions:

- Contrast 1 (combined threshold-and-cessation): [C6, C7] vs. [baseline, C1, C2, C3, C4, C5]. Predicted direction: T2(C6+C7) > T2(baseline + C1-5).
- Contrast 2 (cessation vs. threshold): C7 vs. C6. Predicted direction: T2(C7) > T2(C6).

These two contrasts, together with the Jonckheere-Terpstra ordered test (Section 8.1), constitute the three pre-registered primary tests. All other cross-condition comparisons are treated as exploratory. Family-wise alpha is set at 0.005 one-sided across the three pre-registered tests; Bonferroni correction yields a per-test threshold of alpha approximately 0.00167 one-sided. The strict alpha reflects the foundational nature of the hypothesis: a claim of consciousness-dependent decoherence requires a higher evidential threshold than conventional significance testing provides.

6.2 Null Hypothesis

T2 does not differ systematically across consciousness conditions. The environmental decoherence framework predicts this null. A null result with adequate power constrains, but does not eliminate, the VNW interpretation.

6.3 Alternative Outcomes

- Graded effect: a monotonic ordering across the cognitive-excitation axis (C1 highest decoherence, C7 lowest) would suggest a continuous depth-of-practice effect rather than a single-point cessation effect.
- Single-point effect: C7 substantially exceeds all other conditions, with no clear ordering among C1 to C6. This is the strongest possible signature for a cessation-specific phenomenon.
- Threshold effect: C6 and C7 both exceed C1 to C5 by similar amounts, suggesting the relevant transition is at the MPE threshold rather than at full cessation.
- Null: no systematic differences. This is the prediction of environmental decoherence theory.

SECTION VII

Secondary Experiment: QRNG Test

The secondary experiment uses four quantum random number generators (ComScire PQ32MU, 32 Mbps each) operating continuously throughout each session. The QRNG arm tests whether observer cognitive state correlates with measurable departures from theoretical randomness in quantum-sourced bitstreams.

7.1 Pre-Registered Tests

- Shannon entropy departure from theoretical maximum.
- Autocorrelation at lags 1, 10, 100, 1000.
- Mean deviation across condition windows.
- NIST SP 800-22 statistical test suite (full battery).

7.2 Status of QRNG Arm

The QRNG arm is included as a secondary experiment for two reasons: (i) the consciousness-RNG literature (PEAR, Global Consciousness Project, and subsequent meta-analyses) reports small but reproducible-by-claim effects, and the protocol's shielded chamber and rigorous blinding provide the cleanest environment in which to attempt replication; (ii) the QRNG runs continuously throughout each session at minimal incremental cost. The Phase 0 protocol is not powered to detect the small effect sizes reported in the consciousness-RNG literature; the QRNG arm is exploratory and a null result is not interpreted as a refutation of those claims. A positive QRNG result with consistent direction across conditions would, however, be reportable independently of the T2 arm.

SECTION VIII

Data Analysis and Statistical Framework

8.1 Pre-Registered Primary Family

The pre-registered primary family consists of three tests, evaluated together under a family-wise alpha of 0.005 one-sided with Bonferroni correction (per-test alpha approximately 0.00167 one-sided):

- Test 1, Jonckheere-Terpstra ordered test on the pre-registered consciousness condition ordering (C1, C2, C3, C4, C5, C6, C7). Appropriate because the conditions are ordered by hypothesised cognitive-excitation level and the directional prediction is monotonic across the axis. More powerful than ANOVA when the alternative hypothesis is ordered, as it is here.
- Test 2, Planned Contrast 1: [C6, C7] vs. [baseline, C1-5], one-sided, T2(C6+C7) > T2(baseline + C1-5).
- Test 3, Planned Contrast 2: C7 vs. C6, one-sided, T2(C7) > T2(C6).

All other cross-condition comparisons are treated as exploratory. The strict family-wise alpha (0.005) is chosen because the hypothesis under test, a consciousness-dependent shift in T2, carries unusually high prior implausibility relative to conventional psychology or biology effects; the evidential threshold for accepting a positive result must be correspondingly higher.

8.2 Bayesian Parallel Track

In parallel with the frequentist tests in Section 8.1, each pre-registered test is also evaluated under a Bayesian framework. Bayes factors (BF) are reported for each of the three primary tests with the following pre-registered thresholds:

- BF > 10: strong evidence for the alternative hypothesis (consciousness-dependent T2 shift in the predicted direction).
- BF > 100: decisive evidence for the alternative hypothesis.
- BF < 1/10: strong evidence for the null (no consciousness-dependent shift).
- BF < 1/100: decisive evidence for the null.

Bayesian and frequentist tracks are reported jointly, not as alternatives. A finding is treated as fully pre-registered evidence only when both tracks converge: frequentist significance at the corrected alpha and Bayes factor crossing the corresponding threshold in the same direction. Divergent results (e.g., frequentist significant but BF inconclusive) are reported transparently and flagged as not meeting the pre-registered evidential standard.

8.3 Effect Size

Effect sizes are reported as Cohen's d for each pairwise contrast and as the Jonckheere-Terpstra J statistic for the overall ordering test. The pre-registered target minimum detectable effect for Phase 0 is d = 0.3 on Contrast 1; the sample size required to achieve adequate power at this effect size under the pre-registered alpha is pending final simulation (see Section IX).

8.4 Artifact Analysis

Any T2 anomaly that correlates with an environmental monitor channel is flagged as a candidate artifact and is analyzed separately under pre-registered artifact rules. The artifact analysis is conducted by the independent statistician without access to condition labels.

8.5 Unit of Analysis and Mixed-Effects Model

Because the eligible expert-practitioner pool is small (Section XIII), Phase 0 is a repeated-measures design in which each participant contributes many sessions per condition. These sessions are correlated within participants and are not independent observations; analyzing them as if they were would constitute pseudoreplication and overstate the effective sample size and statistical power.

Accordingly, the pre-registered primary tests (Section 8.1) are evaluated within a linear mixed-effects framework. T2 is the dependent variable; consciousness condition is a fixed effect; and participant is included as a random effect (random intercept, with a random slope across conditions where estimable). This is a within-subjects design: each participant serves as their own control across all seven conditions and the empty-chamber baseline, which removes between-person variance from the contrasts of interest. The planned contrasts and the Jonckheere-Terpstra ordering test are computed on the condition-level estimates from this model. The intraclass correlation (ICC), the proportion of T2 variance attributable to between-participant differences, is reported and used to define the effective sample size.

The unit of statistical inference is therefore the participant-adjusted condition estimate, not the raw session count. Generalization beyond the expert-practitioner sample is not claimed in Phase 0 and is deferred to Phase 1 with a larger participant pool (Section 11.2).

SECTION IX

Power Analysis

Phase 0 is designed around a target minimum detectable effect of d = 0.3 on Contrast 1 ([C6, C7] vs. [baseline, C1-5]) at the pre-registered per-test alpha of 0.00167 one-sided (Bonferroni-corrected from family-wise alpha = 0.005). Achieved power at the planned sample size is pending a final power analysis conducted via permutation simulation against the locked sensor noise model, to be completed and added to the pre-registration before data collection begins. Contrast 2 (C7 vs. C6) is expected to be substantially less well powered than Contrast 1 at the same alpha and sample size; Phase 0 is therefore treated as exploratory for Contrast 2, with adequate power on that contrast deferred to a Phase 1 follow-up using larger samples and the higher-sensitivity transmon sensor.

Phase 0 is explicitly oriented toward the threshold-and-cessation contrast (Contrast 1) and is exploratory for the cessation-vs-threshold contrast (Contrast 2). Small-effect claims from prior consciousness-RNG literature (typical d ~ 0.05 to 0.10) are out of scope for Phase 0; a null result on the QRNG arm does not refute them.

Because Phase 0 is a clustered, repeated-measures design (Section 8.5), the final power analysis treats the within-participant intraclass correlation (ICC) as an explicit parameter. Achieved power depends jointly on the effect size, the number of participants, the sessions per participant, and the ICC; the pre-registered simulation evaluates power across a plausible pre-registered ICC range, with the value confirmed against pilot data before lodging. Power figures that assume independent sessions are not reported, as they would overstate the design's sensitivity.

SECTION X

Stopping Rules

The protocol pre-registers two classes of stopping rules: analytic stopping (criteria for terminating data collection) and operational stopping (criteria for terminating an in-progress session).

10.1 Analytic Stopping

- Data collection proceeds to the pre-registered sample size (50 sessions/condition for the meditator-dependent conditions; 50 sessions for baseline; specified minimums for C2). No interim analyses are conducted.
- If the chamber requires a full re-commissioning event during data collection (post-relocation, post-mechanical-shock, or post-panic-bar event with anomaly), all data collected after the triggering event are analyzed as a separate batch and are not pooled with prior data.

10.2 Operational Stopping

- Any safety threshold breach (O_2 < 19.5%, observer distress, equipment failure with safety implications) terminates the session immediately.
- Panic-bar activation invalidates the session and triggers chamber reverification.
- EEG signature failing to meet the pre-registered target band for the assigned condition by 5 minutes into the condition window triggers session abort; the session is logged but excluded from that condition's analysis cell.

SECTION XI

Budget, Phasing, and Timeline

11.1 Phase 0, Foundational Test

Parameter	Specification
Budget	~$451,200 (range $420,000 to $470,000) for Full Phase 0 configuration
Location	University lab or dedicated experimental facility with purpose-built shielded chamber
Quantum sensor	NV-center in diamond
EEG	64-channel system, optically isolated, battery-powered with fiber-optic data egress
QRNG	4x quantum-source units, fiber-coupled
Shielding	Full dual-mu-metal + copper + Faraday architecture
Personnel	PI + 1 postdoc + meditator participants (voluntary)
Session structure	Single-condition sessions, 5-minute pre-condition settling buffer, 15-minute condition period (C1 to C6) or 30-minute condition period (C7); see Section 5.5.4
Timeline	12 months
Primary scientific goal	Pre-registered eight-cell test (chamber baseline + C1 to C7) of the von Neumann-Wigner interpretation against environmental decoherence theory, using EEG-verified cognitive state as the independent variable and T2 on an NV-center as the dependent variable. Go / no-go for Phase 1.
Co-product	Sensor-and-shielding characterisation paper documenting residual environmental coupling to T2 under the chamber's full isolation regime. Independently publishable in a metrology venue regardless of consciousness-arm outcome.

Table 7: Phase 0 specifications.

The Phase 0 figure reflects the full engineering scope of the chamber (dual mu-metal, active thermal control, LTS SQUID environmental monitoring, optically-isolated EEG). The computed Bill-of-Materials total is approximately $451,200; the stated range $420,000 to $470,000 covers supplier-quote variance and final integration overhead. Two reduced configurations are available if funding requires staging: a Minimal Phase 0 (~$140K to $210K) using a single mu-metal layer, 1.5 m cube, NV + QRNG only and no SQUID; and a Micro-Pilot (~$2K to $6K) using a commercial QRNG only with no shielded chamber. The Micro-Pilot is also described separately in Appendix A as personal exploratory activity outside the pre-registered protocol. The figures above are stated in USD.

NV-centers are characterised in lab cryostats every day; the engineering required for this protocol is not. The chamber's specifications are dictated entirely by what is required to make the consciousness arm interpretable. The metrology paper is what falls out of doing the consciousness test correctly, not the other way around.

11.2 Phase 1, Conditional Follow-up

Parameter	Specification
Budget	~$1M to $3M (full system including dilution refrigerator, transmon platform, underground deployment, and 2 to 3 years of operation)
Location	Underground laboratory (SURF, SNOLAB, LNGS, or Boulby)
Quantum sensor	Superconducting transmon qubit system
Shielding	Full five-layer isolation with airlock
Participant pool	Larger pool, extended data collection
Session structure	Chained multi-condition sessions; 5-minute pre-condition settling buffer precedes each condition; 15 min per condition for C1 to C6 and 30 min for C7 (identical to Phase 0; see Section 5.5.4)
Timeline	2 to 3 years
Adoption requirement	Phase 1 facility access is allocated through established collaborations and is not realistically available to an unaffiliated researcher. The only viable path to Phase 1 is via a host collaboration that adopts the protocol after Phase 0 results justify scaling up.
Goal	Definitive result with adequate statistical power, including adequate power on the C7-vs-C6 contrast.

Table 8: Phase 1 specifications. The Phase 1 figure reflects realistic full-system cost. The dilution refrigerator alone is approximately $234,000 to $585,000 (cryostat module only); the full transmon sensor platform including integration and shielding is $400K to $800K (Table 3, Option A); underground deployment costs and 2 to 3 years of multi-channel operation drive the total into the $1M to $3M range. Phase 1 is conditional on Phase 0 results and on adoption by an institutional host.

11.3 Cost Comparison

Experiment	Cost
Large Hadron Collider (CERN)	$13.25B
LIGO (gravitational wave observatory)	$1.1B
James Webb Space Telescope	$10B
Typical NIH R01 grant	$2.5M
Typical NSF physics grant	$500K
Silence Experiment Phase 0 (this protocol)	~$451K
Silence Experiment Phase 1 (conditional)	$1M to $3M

Table 9: Cost comparison. Phase 0 of the Silence Experiment is comparable in scale to a typical individual research grant; Phase 1 is comparable to a mid-sized collaborative project.

SECTION XII

Ethical Considerations

The protocol involves human participants in altered states of consciousness, including states of complete cognitive cessation. Ethical considerations include:

- Informed consent: participants must understand that they will be sealed in a shielded chamber and asked to enter deep meditation states for extended periods, with particular attention to the structured exit protocol required for C7 (Section 5.5.6). Informed consent includes disclosure of the speculative nature of the hypothesis, the experimental nature of the protocol, and the participant's right to terminate any session immediately via the panic bar.
- Vetted participants: only experienced meditators with documented ability to reliably enter the relevant states are eligible for C5 to C7 conditions, to minimize psychological risk.
- Safety monitoring: continuous fiber-optic video and audio monitoring throughout each session. Automatic abort on safety threshold breach (Section 10.2).
- Post-session debrief: structured debrief with the participant after every session, particularly after C7 sessions, to confirm safe return to ordinary cognitive state and to log any phenomenological notes.
- IRB approval: the protocol requires Institutional Review Board (or equivalent) approval before data collection begins.

SECTION XIII

Limitations

The protocol has several acknowledged limitations:

- Mechanism agnosticism: the experiment does not propose a mechanism for any consciousness-T2 coupling and cannot distinguish among candidate mechanisms (VNW, Orch-OR, or others) if a positive signal is detected. Discrimination among mechanisms would require follow-up experiments with different parameter dependences.
- T2 as proxy: T2 is an operational proxy for any consciousness-coupling channel, not the canonical VNW observable. A null on T2 does not refute VNW in pure measurement-outcome statistics; it constrains it.
- Participant pool: the eligible participant pool for C5 to C7 conditions is small (worldwide pool of experienced contemplative practitioners with documented MPE/cessation reliability). This is a structural constraint on sample size and on the speed at which Phase 0 can be completed.
- Self-selection: participants are self-selected through prior meditation practice. Generalisability to non-practitioners is not addressed by this protocol.
- Phase 0 power on Contrast 2: Phase 0 is not adequately powered for the cessation-vs-threshold contrast (C7 vs. C6). Phase 1 is required for definitive resolution of that contrast.
- Funding and adoption dependencies: Phase 0 requires ~$451K in funding and access to a suitable lab; Phase 1 is gated on institutional adoption (see Section 11.2).
- Positive-direction inference. A positive result establishes a correlation between EEG-verified cognitive state and T2; because no mechanism links consciousness to transverse coherence and T2 is not the canonical measurement-collapse observable, such a result supports but does not by itself confirm the von Neumann-Wigner interpretation. The inferential gap is symmetric with the null-direction caveat above.
- Direction-of-effect assumption. The predicted sign (longer T2 at C6/C7) is an assumption layered on VNW rather than a derived consequence; a reversed sign is physically conceivable and is treated as a pre-registered disconfirmation (Section 6.1), not reinterpreted after the fact.
- Surface cosmic-ray background (Phase 0). A surface laboratory does not attenuate cosmic-ray muons, which carry diurnal and barometric variation. Phase 0 therefore includes a scintillator coincidence veto as a standard (not optional) channel to flag and exclude muon-coincident T2 events; Phase 1's underground siting removes the background.

SECTION XIV

Future Directions

Phase 0 results determine the path forward:

- Positive result, replicated: Phase 1 with transmon sensor, larger participant pool, and adequately powered C7-vs-C6 contrast. Phase 1 facility access via host collaboration with an underground laboratory.
- Positive result, not yet replicated: independent replication by a non-affiliated laboratory before Phase 1 commitment.
- Null result: the pre-registered upper bound on consciousness-T2 coupling is published. The metrology co-product is published regardless.
- Equivocal result: protocol revision and re-pre-registration before any further data collection.

Independent of Phase 0 outcome, the sensor-and-shielding characterisation paper (the metrology co-product) is publishable in a metrology venue and provides quantified upper bounds on residual coupling channels for any future consciousness-T2 experiment using NV-center sensors.

SECTION XV

Conclusion

This protocol proposes a controlled test of the von Neumann-Wigner interpretation of quantum mechanics using EEG-verified cognitive states as the independent variable and T2 decoherence on an NV-center sensor as the dependent variable, inside a purpose-built five-layer shielded chamber designed to drive every known environmental coupling channel below the sensor's intrinsic T2 limit. The protocol's value is in the specificity of its pre-registered predictions, the rigor of its environmental controls and blinding, and the transparency of its limitations. It is structured as a foundational Phase 0 test (~$451K, NV-center, university lab) followed by a conditional Phase 1 (~$1M to $3M, transmon, underground facility) gated on Phase 0 results and institutional adoption.

Null results establish pre-registered upper bounds on any consciousness-T2 coupling. Positive results require independent replication before being treated as established. Either outcome contributes to a question that has been part of the quantum foundations literature for nearly a century without ever being tested this way.

APPENDIX

Personal Exploratory Activity (Not Part of the Main Protocol)

Independently of this pre-registered protocol, the principal investigator may conduct small-scale exploratory measurements using a commercial quantum random number generator (ComScire PQ4000KU or equivalent) at a personal location, with no shielded chamber, no EEG verification, and no formal blinding. The cost of this activity is approximately $2,000 to $5,000 and is funded personally.

This activity is included in this paper for transparency, not as part of the experimental protocol. It is explicitly flagged as personal exploratory activity, not pre-registered, and not subject to the analysis framework of Section 8. Any results from this activity are not reported as findings of the Silence Experiment and are not pooled with the pre-registered Phase 0 or Phase 1 data.

The activity's only function is to develop the PI's practical familiarity with QRNG operation, data handling, and the statistical signatures of quantum-sourced bitstreams under everyday conditions, in advance of the formal Phase 0 protocol. It is described here so that any reader who encounters the PI's personal notes or informal reports on this activity understands its status: separate from, and not constitutive of, the experiment described in this paper.

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